

MCV4UY - Unit 3 Quiz - Derivatives

Name _____

Date _____

Total
/25

KNOWLEDGE

1. Differentiate the following using different rules. Simply/factor parts b and c only. (NO negative exponents in final answers).

a. $y = \sqrt{x} + \frac{3}{x^{-2}} - 6x^{\frac{5}{3}} + \frac{7}{x^3}$

b. $f(x) = (3x^3 + 4x^2)(2x^4 - 5x)$

(2)

(3)

c. $h(x) = \frac{x^5 - 2x^4 + 3x^3}{4x}$

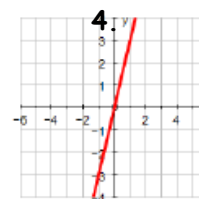
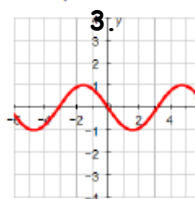
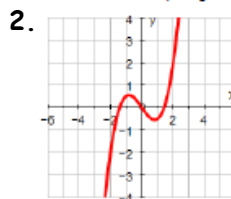
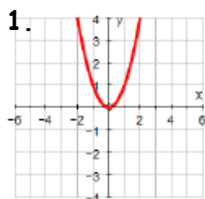
d. $g(x) = \frac{-3x^{-3} - 1}{2x^2 + 1}$

(2)

(3)

2. The graphs in the first row are derivatives. Match them with the graph of their function shown in the second row.

(Graphs of Derivative)



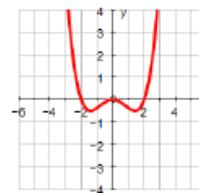
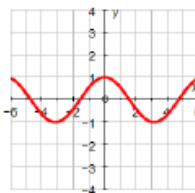
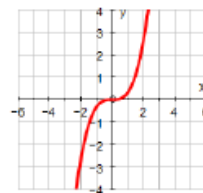
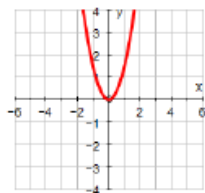
1. _____

2. _____

3. _____

4. _____

(2)



(Graphs of Function)

A.

B.

C.

D.

APPLICATION

3. Find the constants a , b and c so that the graph of $y = ax^2 + bx + c$ contains the point $(-1, 1)$ and is tangent to the line $y = 2x$ at $(0, 0)$.

(3)

4. Solve for a and b in order for $f(x)$ to be both continuous and differentiable at $x=1$.

$$f(x) = \begin{cases} x^2 + 2, & x \leq 1 \\ a\left(x - \frac{1}{x}\right) + b, & x > 1 \end{cases}$$

(3)

COMMUNICATION

6. Discuss the relationship between continuity and differentiability of a function. Give examples of 4 ways for a function not to be differentiable.

(3)

7. Use the graph of $f(x)$ shown on the right.

a. Where is $f'(x) = 0$?

(1)

b. Where is $f'(x) > 0$?

(1)

c. Where is $f'(x) < 0$?

(1)

d. On the same graph, draw a possible sketch of $f'(x)$.

(1)

